

## Spike & QEMU Research

### Spike

Spike is the official RISC-V ISA simulator.

Purpose:

- Test RISC-V programs
- Validate ISA implementation
- Educational research

Advantages:

- Lightweight
- Accurate ISA simulation
- Official reference implementation

Limitations:

- Does not emulate complete hardware systems
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### QEMU

QEMU is an open-source machine emulator and virtualizer.

Purpose:

- Emulate complete computer systems
- Run operating systems for different architectures
- Test embedded software

Advantages:

- Supports multiple architectures
  - Fast emulation
  - Widely used in embedded development
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### Spike vs QEMU

| Spike                                               | QEMU                        |
|-----------------------------------------------------|-----------------------------|
| ISA Simulator                                       | Full System Emulator        |
| Lightweight                                         | Complete hardware emulation |
| Education & ISA Testing OS and Embedded Development |                             |

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## **Why They Are Important**

### **Spike**

- ISA verification
- Learning RISC-V
- Instruction testing

### **QEMU**

- Embedded Linux
- Firmware development
- Hardware simulation

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## **Applications**

- Processor research
- Embedded systems
- Computer architecture education
- Operating system development

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## **Key Learning**

- Learned the purpose of Spike and QEMU.
- Understood the difference between ISA simulation and full-system emulation.
- Explored their role in RISC-V processor development and embedded systems research.